Group II Row IV from $\mathfrak{C}_1$: your derivation, corrected

Corrections (blue = new, red = removed). Every other step is correct as written; the checks are in `checks.py`.

C1 Last equation, the log $\frac{\vee}{k^+}$ H_4 line: the sign at the start of the line was missing. It is $+$; without it the split does not add up, and LaTeX prints the line as a product with the previous one.

C2 The $\frac{1}{\epsilon}$ is the collinear (infrared) divergence of the z integral (large $|z-y|$), not an ultraviolet one. Only the wording changes.

C3 The upper limit $\frac{k^+}{k^+ + \Lambda} \to 1$, in the plus-prescribed integral and in the linear-polarization line, is the limit $\Lambda \to 0$: it drops $O(\frac{\Lambda}{k^+})$.

C4 With $d_\perp = 2 - \epsilon$ kept: the two H_4 lines carry $\frac{d_\perp}{2}$, like the $y' = y$ subtraction line. Then the last equation equals the one before it exactly.

Checked and correct:

- the prefactor $4 \cdot \frac{g^2}{2\pi^2 P} \cdot \frac{g^2\, p^+ k^+}{8\pi^2 P^3} = \frac{g^4 p^+ k^+}{4\pi^4 P^4}$ $(P = p^+ + k^+)$;
- the w, w' integrals: Jacobian $(P/k^+)^2$ each, $w - z = \frac{P}{k^+}(y - z)$, phase $e^{-ik(y'-y)P/k^+}$;
- the i, j contraction and the bracket with $\frac{1}{k^+}$ (symbolic, and index by index in $d = 2, 3$);
- the Wilson-line reduction to $N_c[U(y') - U(x')][U(y) - U(x)]$;
- the z integral: trace $-\pi[\frac{2}{\epsilon} + 2\gamma_E + \log\frac{(y'-y)^2\mu^2}{4}]$, traceless part $-\pi\frac{(y'-y)^m (y'-y)^{m'}}{(y'-y)^2}$;
- the drop-out of the $\frac{2}{k^+}$ terms;
- the change of variables $\xi = \frac{k^+}{p^+ + k^+}$, with its limits and sign;
- the blue $\log\frac{k^+}{\Lambda} + \log(\frac{\vee}{k^+} - 1)$, which is exact with your plus prescription.

$$\frac{d^3 N_{\rm NLO}}{d^3 k} = \frac{4}{(2\pi)^3}\int_\Lambda^{\vee - k^+}\frac{dp^+}{2\pi}\int_{z,w,w'} e^{-ik(w'-w)}\int_{y,y'}\Big[\mathbb{A}^{\dagger\,(1)\,e'}_{k'}(p^+ + k^+, -y')U^{\dagger\,e'c'}(y')U^{d'b}(z)\bar{\mathbb{C}}^{\dagger\,d'ac'}_{1jik'}(p^+ + k^+, y' - w'; p^+, w' - z)$$
$$- \bar{\mathbb{A}}^{\dagger\,(1)\,c'}_{k'}(p^+ + k^+, -y')\bar{\mathbb{C}}^{\dagger\,d'ac'}_{1jik'}(p^+ + k^+, y' - w'; p^+, w' - z)U^{d'b}(z)\Big] \times \Big[\bar{\mathbb{C}}^{dac}_{1jik}(p^+ + k^+, y - w; p^+, w - z)$$
$$U^{bd}(z)U^{ce}(y)\mathbb{A}^{(1)\,e}_k(p^+ + k^+, -y) - U^{bd}(z)\bar{\mathbb{C}}^{dac}_{1jik}(p^+ + k^+, y - w; p^+, w - z)\bar{\mathbb{A}}^{(1)\,c}_k(p^+ + k^+, -y)\Big]$$

$$\frac{d^3 N_{\rm NLO}}{d^3 k} = \frac{1}{(2\pi)^3}\frac{g^4}{4\pi^4}\int_\Lambda^{\vee - k^+}\frac{dp^+}{2\pi}\int_{z,w,w',x'} e^{-ik(w'-w)}\int_{x,y,y'}\frac{(w-z)^m (x-y)^k (w'-z)^{m'} (x'-y')^{k'}}{(w-z)^2 (x-y)^2 (w'-z)^2 (x'-y')^2}\frac{p^+ k^+}{(p^+ + k^+)^4}$$
$$\delta^{(2)}\left[y - w + \frac{p^+}{(p^+ + k^+)}(w - z)\right] \times \delta^{(2)}\left[y' - w' + \frac{p^+}{(p^+ + k^+)}(w' - z)\right] \times$$
$$\left[\delta_{km}\,\delta_{ij} - \frac{(p^+ + k^+)}{p^+}\,\delta_{jm}\,\delta_{ik} - \frac{(p^+ + k^+)}{k^+}\,\delta_{im}\,\delta_{kj}\right] \times \left[\delta_{k'm'}\,\delta_{ij} - \frac{(p^+ + k^+)}{p^+}\,\delta_{jm'}\,\delta_{ik'} - \frac{(p^+ + k^+)}{k^+}\,\delta_{im'}\,\delta_{k'j}\right]$$
$$\left[f^{c'd'a}\,U^{e'c'}(y')\,U^{d'b}(z)\,\rho^{e'}(x') - f^{c'd'a}\,U^{bd'}(z)\,U^{c'e'}(x')\,\rho^{e'}(x')\right] \times \left[f^{cda}\,U^{ec}(y)\,U^{db}(z)\,\rho^e(x) - f^{cda}\,U^{bd}(z)\,U^{ce}(x)\,\rho^e(x)\right] \quad (1)$$

Integrating w and w',

$$\frac{d^3 N_{\rm NLO}}{d^3 k} = \frac{1}{(2\pi)^3}\frac{g^4}{4\pi^4}\frac{1}{k^+}\int_\Lambda^{\vee - k^+}\frac{dp^+}{2\pi}\int_z e^{-ik\,(y'-y)\,\frac{(k^+ + p^+)}{k^+}}\int_{x,y,y',x'}\frac{(y-z)^m (x-y)^k (y'-z)^{m'} (x'-y')^{k'}}{(y-z)^2 (x-y)^2 (y'-z)^2 (x'-y')^2}$$
$$\left[\frac{\delta_{km}\delta_{k'm'}\,d_\perp\,p^+}{(k^+ + p^+)^2} + \frac{2}{k^+}\,(\delta_{mk'}\delta_{m'k} - \delta_{m'k'}\delta_{km}) + \frac{p^+}{k^{+\,2}}\,\delta_{mm'}\delta_{kk'} + \frac{1}{p^+}\,\delta_{mm'}\delta_{kk'}\right] \times$$
$$\left[f^{c'd'a}\,U^{e'c'}(y')\,U^{d'b}(z)\,\rho^{e'}(x') - f^{c'd'a}\,U^{bd'}(z)\,U^{c'e'}(x')\,\rho^{e'}(x')\right] \times \left[f^{cda}\,U^{ec}(y)\,U^{db}(z)\,\rho^e(x) - f^{cda}\,U^{bd}(z)\,U^{ce}(x)\,\rho^e(x)\right] \quad (2)$$

Reducing Wilson lines, we get

$$\frac{d^3 N_{\rm NLO}}{d^3 k} = \frac{1}{(2\pi)^3}\frac{g^4}{4\pi^4}\frac{N_c}{k^+}\int_\Lambda^{\vee - k^+}\frac{dp^+}{2\pi}\int_z e^{-ik\,(y'-y)}\,e^{-ik\,(y'-y)\,\frac{p^+}{k^+}}\int_{x,y,y',x'}\frac{(y-z)^m (x-y)^k (y'-z)^{m'} (x'-y')^{k'}}{(y-z)^2 (x-y)^2 (y'-z)^2 (x'-y')^2}$$

$$\left[\frac{\delta_{km}\delta_{k'm'}\,d_\perp\,p^+}{(k^++p^+)^2}+\frac{2}{k^+}\,(\delta_{mk'}\delta_{m'k}-\delta_{m'k'}\delta_{km})+\frac{p^+}{k^{+\,2}}\,\delta_{mm'}\delta_{kk'}+\frac{1}{p^+}\,\delta_{mm'}\delta_{kk'}\right]\times$$
$$\left[U^{e'c}(y')-U^{e'c}(x')\right]\times\left[U^{ec}(y)-U^{ce}(x)\right]\,\rho^{e'}(x')\rho^{e}(x) \tag{3}$$

We are performing z integral and using $\overline{MS}$ and $d_\perp=2-\epsilon$, we get

$$\int_z\frac{(y-z)^m\,(y'-z)^{m'}}{(y-z)^2\,(y'-z)^2}=-\frac{\pi}{2}\left[\frac{2}{\epsilon}+2\gamma_E+\log\left[\frac{(y'-y)^2\,\mu^2_{\overline{MS}}}{4}\right]\right]\delta^{mm'}\;-\pi\,\frac{(y'-y)^m\,(y'-y)^{m'}}{(y'-y)^2} \tag{4}$$

This gives us

$$\begin{aligned}\frac{d^3N_{\rm NLO}}{d^3k}=&-\frac{1}{(2\pi)^3}\,\frac{g^4\,d_\perp}{4\pi^3}\,\frac{N_c}{k^+}\int_\Lambda^{\vee-k^+}\frac{dp^+}{2\pi}\int_{x,y,y',x'}e^{-ik\,(y'-y)\,\frac{p^++k^+}{k^+}}\,\frac{(x-y)\cdot(x'-y')}{(x-y)^2\,(x'-y')^2}\\
&\left[\frac{p^+}{(k^++p^+)^2}+\frac{p^+}{k^{+\,2}}+\frac{1}{p^+}\right]\times\left[\frac{1}{\epsilon}+\gamma_E+\frac{1}{2}\,\log\left[\frac{(y'-y)^2\,\mu^2_{\overline{MS}}}{4}\right]\right]\\
&\left[U^{e'c}(y')-U^{e'c}(x')\right]\times\left[U^{ec}(y)-U^{ce}(x)\right]\,\rho^{e'}(x')\rho^{e}(x)\\
&-\frac{1}{(2\pi)^3}\,\frac{g^4}{4\pi^3}\,\frac{N_c}{k^+}\int_\Lambda^{\vee-k^+}\frac{dp^+}{2\pi}\int_{x,y,y',x'}e^{-ik\,(y'-y)\,\frac{p^++k^+}{k^+}}\,\frac{(x-y)^k\,(x'-y')^{k'}\,(y'-y)^m\,(y'-y)^{m'}}{(x-y)^2\,(x'-y')^2\,(y'-y)^2}\\
&\left[\frac{\delta_{km}\delta_{k'm'}\,d_\perp\,p^+}{(k^++p^+)^2}+\frac{2}{k^+}\,(\delta_{mk'}\delta_{m'k}-\delta_{m'k'}\delta_{km})+\frac{p^+}{k^{+\,2}}\,\delta_{mm'}\delta_{kk'}+\frac{1}{p^+}\,\delta_{mm'}\delta_{kk'}\right]\times\\
&\left[U^{e'c}(y')-U^{e'c}(x')\right]\times\left[U^{ec}(y)-U^{ce}(x)\right]\,\rho^{e'}(x')\rho^{e}(x)\end{aligned} \tag{5}$$

Simplifying, we get

$$\begin{aligned}\frac{d^3N_{\rm NLO}}{d^3k}=&-\frac{1}{(2\pi)^3}\,\frac{g^4\,d_\perp}{4\pi^3}\,\frac{N_c}{k^+}\int_\Lambda^{\vee-k^+}\frac{dp^+}{2\pi}\int_{x,y,y',x'}e^{-ik\,(y'-y)\,\frac{p^++k^+}{k^+}}\,\frac{(x-y)\cdot(x'-y')}{(x-y)^2\,(x'-y')^2}\left[\frac{p^+}{(k^++p^+)^2}+\frac{p^+}{k^{+\,2}}+\frac{1}{p^+}\right]\times\\
&\left[\frac{1}{\epsilon}+\gamma_E+\frac{1}{2}\,\log\left[\frac{(y'-y)^2\,\mu^2_{\overline{MS}}}{4}\right]\right]\left[U^{e'c}(y')-U^{e'c}(x')\right]\times\left[U^{ec}(y)-U^{ce}(x)\right]\,\rho^{e'}(x')\rho^{e}(x)\\
&-\frac{1}{(2\pi)^3}\,\frac{g^4}{4\pi^3}\,\frac{N_c}{k^+}\int_\Lambda^{\vee-k^+}\frac{dp^+}{2\pi}\int_{x,y,y',x'}e^{-ik\,(y'-y)\,\frac{p^++k^+}{k^+}}\left[\frac{(x-y)\cdot(y'-y)\,(x'-y')\cdot(y'-y)}{(x-y)^2\,(x'-y')^2\,(y'-y)^2}\,\frac{d_\perp\,p^+}{(k^++p^+)^2}\right.\\
&\left.+\frac{(x-y)\cdot(x'-y')}{(x-y)^2\,(x'-y')^2}\,\left[\frac{p^+}{k^{+\,2}}+\frac{1}{p^+}\right]\right]\times\left[U^{e'c}(y')-U^{e'c}(x')\right]\times\left[U^{ec}(y)-U^{ce}(x)\right]\,\rho^{e'}(x')\rho^{e}(x)\end{aligned} \tag{6}$$

Using $\xi=\frac{k^+}{(p^++k^+)}$, we get

$$\begin{aligned}\frac{d^3N_{\rm NLO}}{d^3k}=&\frac{1}{(2\pi)^3}\,\frac{g^4\,d_\perp}{4\pi^3}\,\frac{N_c}{k^+}\int_{\frac{k^+}{k^++\Lambda}}^{\frac{k^+}{\vee}}\frac{d\xi}{2\pi}\int_{x,y,y',x'}e^{-ik\,(y'-y)\,\frac{1}{\xi}}\,\frac{(x-y)\cdot(x'-y')}{(x-y)^2\,(x'-y')^2}\left[\xi\,(1-\xi)+\frac{(1-\xi)}{\xi}+\frac{\xi}{(1-\xi)}\right]\frac{1}{\xi^2}\\
&\left[\frac{1}{\epsilon}+\gamma_E+\frac{1}{2}\,\log\left[\frac{(y'-y)^2\,\mu^2_{\overline{MS}}}{4}\right]\right]\left[U^{e'c}(y')-U^{e'c}(x')\right]\times\left[U^{ec}(y)-U^{ce}(x)\right]\,\rho^{e'}(x')\rho^{e}(x)\\
&+\frac{1}{(2\pi)^3}\,\frac{g^4}{4\pi^3}\,\frac{N_c}{k^+}\int_{\frac{k^+}{k^++\Lambda}}^{\frac{k^+}{\vee}}\frac{d\xi}{2\pi}\int_{x,y,y',x'}e^{-ik\,(y'-y)\,\frac{1}{\xi}}\,\frac{1}{\xi^2}\left[\frac{(x-y)\cdot(y'-y)\,(x'-y')\cdot(y'-y)}{(x-y)^2\,(x'-y')^2\,(y'-y)^2}\,d_\perp\,\xi(1-\xi)\right.\\
&\left.+\frac{(x-y)\cdot(x'-y')}{(x-y)^2\,(x'-y')^2}\,\left[\frac{(1-\xi)}{\xi}+\frac{\xi}{(1-\xi)}\right]\right]\times\left[U^{e'c}(y')-U^{e'c}(x')\right]\times\left[U^{ec}(y)-U^{ce}(x)\right]\,\rho^{e'}(x')\rho^{e}(x)\end{aligned} \tag{7}$$

We write it as follows:

$$\begin{aligned}\frac{d^3N_{\rm NLO}}{d^3k}=&\frac{g^2}{8\pi^2}\,\frac{1}{(2\pi)^3}\,\frac{g^2}{2\pi^2}\,\frac{1}{k^+}\int_{\frac{k^+}{\vee}}^{\frac{k^+}{k^++\Lambda}}\frac{d\xi}{\xi^2}\int_{x,y,x',y'}e^{-ik\cdot(y'-y)\,\frac{1}{\xi}}\left\{\mathcal{P}_{gg}(\xi)\,\frac{(x-y)\cdot(x'-y')}{(x-y)^2\,(x'-y')^2}\right.\\
&\left[d_\perp\,\left[-\frac{1}{\epsilon}-\gamma_E-\frac{1}{2}\,\log\left(\frac{(y'-y)^2\,\mu^2_{\overline{MS}}}{4}\right)\right]-1\right]+2\,N_c\,\xi(1-\xi)\left[\frac{(x-y)\cdot(x'-y')}{(x-y)^2\,(x'-y')^2}\right.\\
&\left.\left.-\frac{(x-y)\cdot(y'-y)\,(x'-y')\cdot(y'-y)}{(x-y)^2\,(x'-y')^2\,(y'-y)^2}\,d_\perp\right]\right\}\times\left[U^{e'c}(y')-U^{e'c}(x')\right]\times\left[U^{ec}(y)-U^{ce}(x)\right]\,\rho^{e'}(x')\rho^{e}(x)\end{aligned} \tag{8}$$

The pole is present in the following:

$$\begin{aligned}\frac{d^3N_{\rm pole}}{d^3k}=&-\frac{g^2}{8\pi^2}\,\frac{1}{(2\pi)^3}\,\frac{g^2}{2\pi^2}\,\frac{d_\perp}{k^+}\,\frac{1}{\epsilon}\int_{\frac{k^+}{\vee}}^{\frac{k^+}{k^++\Lambda}}\frac{d\xi}{\xi^2}\int_{x,y,x',y'}e^{-ik\cdot(y'-y)\,\frac{1}{\xi}}\,\mathcal{P}_{gg}(\xi)\,\frac{(x-y)\cdot(x'-y')}{(x-y)^2\,(x'-y')^2}\\
&\left[U^{e'c}(y')-U^{e'c}(x')\right]\times\left[U^{ec}(y)-U^{ce}(x)\right]\,\rho^{e'}(x')\rho^{e}(x)\end{aligned} \tag{9}$$

where $\mathcal{P}_{gg}(\xi)=2\,N_c\left[\xi\,(1-\xi)+\frac{(1-\xi)}{\xi}+\frac{\xi}{(1-\xi)}\right]$ is the gluon - gluon splitting function. Since we have handled the UV divergencewe have isolated the collinear (infrared) divergence: the z integral diverges at large $|z-y|$, not at $z\to y$, where its integrand

goes only like $1/|z-y|$, now we can safely use plus prescription to integrate wrt ξ. Here $\int \frac{d\xi}{\xi^2}\frac{\xi}{(1-\xi)_+}G \equiv \int \frac{d\xi}{\xi(1-\xi)}[G(\xi)-G(1)]$ (your prescription). With it, $\int_{k^+/\vee}^{k^+/(k^++\Lambda)} \frac{d\xi}{\xi(1-\xi)} = \log\frac{k^+}{\Lambda} + \log(\frac{\vee}{k^+}-1)$ exactly, which is the blue line below.

$$
\begin{aligned}
\frac{d^3N_{\rm NLO}}{d^3k} &= \frac{g^2}{4\pi^2}\,\frac{N_c}{(2\pi)^3}\,\frac{g^2}{2\pi^2}\,\frac{1}{k^+}\int_{x,y,x',y'}\frac{(x-y)\cdot(x'-y')}{(x-y)^2\,(x'-y')^2}\left[-\frac{d_\perp}{\epsilon}-d_\perp\,\gamma_E-\frac{d_\perp}{2}\,\log\left(\frac{(y'-y)^2\,\mu_{\overline{MS}}^2}{4}\right)-1\right]\\
&\left[\int_{\frac{k^+}{\vee}}^1\frac{d\xi}{\xi^2}\,e^{-ik\cdot(y'-y)\,\frac{1}{\xi}}\left[\xi\,(1-\xi)+\frac{(1-\xi)}{\xi}+\frac{\xi}{(1-\xi)_+}\right]\right]\times\left[U^{e'c}(y')-U^{e'c}(x')\right]\times\left[U^{ec}(y)-U^{ce}(x)\right]\,\rho^{e'}(x')\rho^e(x)\\
&+\frac{g^2}{4\pi^2}\,\frac{1}{(2\pi)^3}\,\frac{g^2}{2\pi^2}\,\frac{N_c}{k^+}\left[\log\left(\frac{k^+}{\Lambda}\right)+\log\left(\frac{\vee}{k^+}-1\right)\right]\int_{x,y,x',y'}e^{-ik\cdot(y'-y)}\,\frac{(x-y)\cdot(x'-y')}{(x-y)^2\,(x'-y')^2}\left[-\frac{d_\perp}{\epsilon}-d_\perp\,\gamma_E\right.\\
&\left.-\frac{d_\perp}{2}\,\log\left(\frac{(y'-y)^2\,\mu_{\overline{MS}}^2}{4}\right)-1\right]\left[U^{e'c}(y')-U^{e'c}(x')\right]\times\left[U^{ec}(y)-U^{ce}(x)\right]\,\rho^{e'}(x')\rho^e(x)\\
&+\frac{g^2}{4\pi^2}\,\frac{1}{(2\pi)^3}\,\frac{g^2}{2\pi^2}\,\frac{N_c}{k^+}\int_{\frac{k^+}{\vee}}^{\frac{k^+}{k^++\Lambda}}d\xi\,\frac{(1-\xi)}{\xi}\int_{x,y,x',y'}e^{-ik\cdot(y'-y)\,\frac{1}{\xi}}\left[\frac{(x-y)\cdot(x'-y')}{(x-y)^2\,(x'-y')^2}-\frac{(x-y)\cdot(y'-y)\,(x'-y')\cdot(y'-y)}{(x-y)^2\,(x'-y')^2\,(y'-y)^2}\,d_\perp\right]\times\\
&\left[U^{e'c}(y')-U^{e'c}(x')\right]\times\left[U^{ec}(y)-U^{ce}(x)\right]\,\rho^{e'}(x')\rho^e(x)
\end{aligned}
\tag{10}
$$

In the first line the upper limit $\frac{k^+}{k^++\Lambda}$ of the plus-prescribed integral is set to 1. This drops $\int_{k^+/(k^++\Lambda)}^1 \frac{d\xi}{\xi(1-\xi)}[G(\xi)-G(1)] = O(\frac{\Lambda}{k^+})$, i.e. it is the limit $\Lambda\to 0$. The same holds for the upper limit of the linear-polarization line, $\frac{k^+}{k^++\Lambda}\to 1$, in the next equation.

We can split this as :

$$
\begin{aligned}
\frac{d^3N_{\rm NLO}}{d^3k} &= \frac{g^2}{4\pi^2}\,\frac{N_c}{(2\pi)^3}\,\frac{g^2}{2\pi^2}\,\frac{1}{k^+}\int_{x,y,x',y'}\frac{(x-y)\cdot(x'-y')}{(x-y)^2\,(x'-y')^2}\left[-\frac{d_\perp}{\epsilon}-d_\perp\,\gamma_E-\frac{d_\perp}{2}\,\log\left(\frac{(y'-y)^2\,\mu_{\overline{MS}}^2}{4}\right)-1\right]\\
&\left[\int_{\frac{k^+}{\vee}}^1\frac{d\xi}{\xi^2}\,e^{-ik\cdot(y'-y)\,\frac{1}{\xi}}\left[\xi\,(1-\xi)+\frac{(1-\xi)}{\xi}+\xi+\frac{\xi^2}{(1-\xi)_+}\right]\right]\times\left[U^{e'c}(y')-U^{e'c}(x')\right]\times\left[U^{ec}(y)-U^{ce}(x)\right]\,\rho^{e'}(x')\rho^e(x)\\
&-\frac{g^2}{4\pi^2}\,\frac{N_c}{(2\pi)^3}\,\frac{g^2}{2\pi^2}\,\frac{1}{k^+}\int_{x,y,x',y'}\frac{(x-y)\cdot(x'-y)}{(x-y)^2\,(x'-y)^2}\left[-\frac{d_\perp}{2}\,\log\left(\frac{(y'-y)^2\,\mu_{\overline{MS}}^2}{4}\right)\right]\\
&\left[\int_{\frac{k^+}{\vee}}^1\frac{d\xi}{\xi^2}\,e^{-ik\cdot(y'-y)\,\frac{1}{\xi}}\left[\frac{(1-\xi)}{\xi}\right]\right]\times\left[U^{e'c}(y)-U^{e'c}(x')\right]\times\left[U^{ec}(y)-U^{ce}(x)\right]\,\rho^{e'}(x')\rho^e(x)\\
&-\frac{d_\perp}{2}\,\frac{1}{(2\pi)^3}\,\frac{g^4\,N_c}{8\pi^5}\,\frac{1}{k^+}\left(1-\frac{k^+}{\vee}\right)\int_{x,x',z,w,w'}e^{-ik\cdot(w'-w)}\,\frac{(w-z)\cdot(w'-z)}{(w-z)^2\,(w'-z)^2}\,\frac{(x-z)\cdot(x'-z)}{(x-z)^2\,(x'-z)^2}\\
&\left[U^{e'c}(z)-U^{e'c}(x')\right]\times\left[U^{ec}(z)-U^{ce}(x)\right]\,\rho^{e'}(x')\rho^e(x)\\
&+\frac{g^2}{4\pi^2}\,\frac{1}{(2\pi)^3}\,\frac{g^2}{2\pi^2}\,\frac{N_c}{k^+}\log\left(1-\frac{k^+}{\vee}\right)\int_{x,y,x',y'}e^{-ik\cdot(y'-y)}\,\frac{(x-y)\cdot(x'-y')}{(x-y)^2\,(x'-y')^2}\left[-\frac{d_\perp}{\epsilon}-d_\perp\,\gamma_E\right.\\
&\left.-\frac{d_\perp}{2}\,\log\left(\frac{(y'-y)^2\,\mu_{\overline{MS}}^2}{4}\right)-1\right]\left[U^{e'c}(y')-U^{e'c}(x')\right]\times\left[U^{ec}(y)-U^{ce}(x)\right]\,\rho^{e'}(x')\rho^e(x)\\
&+\frac{g^2}{4\pi^2}\,\frac{1}{(2\pi)^3}\,\frac{g^2}{2\pi^2}\,\frac{N_c}{k^+}\int_{\frac{k^+}{\vee}}^{\frac{k^+}{k^++\Lambda}\to 1}d\xi\,\frac{(1-\xi)}{\xi}\int_{x,y,x',y'}e^{-ik\cdot(y'-y)\,\frac{1}{\xi}}\left[\frac{(x-y)\cdot(x'-y')}{(x-y)^2\,(x'-y')^2}-\frac{(x-y)\cdot(y'-y)\,(x'-y')\cdot(y'-y)}{(x-y)^2\,(x'-y')^2\,(y'-y)^2}\,d_\perp\right]\times\\
&\left[U^{e'c}(y')-U^{e'c}(x')\right]\times\left[U^{ec}(y)-U^{ce}(x)\right]\,\rho^{e'}(x')\rho^e(x)\\
&+\frac{g^2}{4\pi^2}\,\frac{1}{(2\pi)^3}\,\frac{g^2}{2\pi^2}\,\frac{N_c}{k^+}\log\left(\frac{\vee}{k^+}\right)\int_{x,y,x',y'}e^{-ik\cdot(y'-y)}\,\frac{(x-y)\cdot(x'-y')}{(x-y)^2\,(x'-y')^2}\left[-\frac{d_\perp}{\epsilon}-d_\perp\,\gamma_E\right.\\
&\left.-\frac{d_\perp}{2}\,\log\left(\frac{(y'-y)^2\,\mu_{\overline{MS}}^2}{4}\right)-1\right]\left[U^{e'c}(y')-U^{e'c}(x')\right]\times\left[U^{ec}(y)-U^{ce}(x)\right]\,\rho^{e'}(x')\rho^e(x)\\
&-\frac{g^2}{4\pi^2}\,\frac{1}{(2\pi)^3}\,\frac{g^2}{2\pi^2}\,\frac{N_c}{k^+}\log\left(\frac{\vee}{k^+}\right)\int_{x,y,x',y'}e^{-ik\cdot(y'-y)}\,\frac{(x-y)\cdot(x'-y')}{(x-y)^2\,(x'-y')^2}\left[-\frac{d_\perp}{\epsilon}-d_\perp\,\gamma_E\right.\\
&\left.-\frac{d_\perp}{2}\,\log\left(\frac{(y'-y)^2\,\mu_{\overline{MS}}^2}{4}\right)-1\right]\left[U^{e'c}(y')-U^{e'c}(x')\right]\times\left[U^{ec}(y)-U^{ce}(x)\right]\,\rho^{e'}(x')\rho^e(x)\\
&+\frac{d_\perp}{2}\,\frac{1}{(2\pi)^3}\,\frac{g^4\,N_c}{8\pi^5}\,\frac{1}{k^+}\log\left(\frac{\vee}{k^+}\right)\int_{x,x',z,w,w'}e^{-ik\cdot(w'-w)}\,\frac{(w-z)\cdot(w'-z)}{(w-z)^2\,(w'-z)^2}\,\frac{(x-z)\cdot(x'-z)}{(x-z)^2\,(x'-z)^2}\\
&\left[U^{e'c}(z)-U^{e'c}(x')\right]\times\left[U^{ec}(z)-U^{ce}(x)\right]\,\rho^{e'}(x')\rho^e(x)\\
&+\frac{1}{(2\pi)^3}\,\frac{g^4}{8\pi^4}\,\frac{N_c}{k^+}\log\left(\frac{k^+}{\Lambda}\right)\int_{x,y,x',y'}e^{-ik\cdot(y'-y)}\,\frac{(x-y)\cdot(x'-y')}{(x-y)^2\,(x'-y')^2}\left[-\frac{d_\perp}{\epsilon}-d_\perp\,\gamma_E\right.\\
&\left.-\frac{d_\perp}{2}\,\log\left(\frac{(y'-y)^2\,\mu_{\overline{MS}}^2}{4}\right)-1\right]\left[U^{e'c}(y')-U^{e'c}(x')\right]\times\left[U^{ec}(y)-U^{ce}(x)\right]\,\rho^{e'}(x')\rho^e(x)
\end{aligned}
\tag{11}
$$

Check of the split. The two lines with $\pm\log\frac{\vee}{k^+}$ times the bracket cancel. Your $y'=y$ subtraction line gives, after the y'

integral, $-\frac{d_\perp}{2}\left[\log\frac{\vee}{k^+}-(1-\frac{k^+}{\vee})\right]H_4$. Together with the two H_4 lines, $-\frac{d_\perp}{2}(1-\frac{k^+}{\vee})H_4$ and $+\frac{d_\perp}{2}\log\frac{\vee}{k^+}H_4$, it adds up to zero. Hence the last equation equals the one before it, exactly at $d_\perp = 2-\epsilon$. Here H_4 is the $\int_{x,x',z,w,w'}$ structure of those lines.